

MoodleQuest: Gamified Engagement for Asynchronous Learning

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Abstract—MoodleQuest is a gamified companion system designed to improve student participation in asynchronous Moodle courses while preserving transparency and institutional trust. Prior studies indicate that timely engagement cues, visible rewards, and structured incentives can enhance learner motivation and course persistence. Research on online learning further shows that near-real-time feedback and multi-factor analytics support higher completion rates and deeper engagement. Guided by these findings, MoodleQuest integrates quests, experience points, badges, leaderboards, virtual pets, and progress indicators to encourage consistent study habits and active interaction with course materials. The system connects to Moodle using secure single sign-on, role-based access control, and web-service integration, enabling data synchronization without duplicate accounts. A Next.js front end and FastAPI middleware provide dashboards that visualize engagement heatmaps, streak patterns, and early warnings for at-risk students. Initial walkthroughs indicate that daily quests and streak-based rewards help learners better understand their effort and maintain learning momentum. Planned pilot studies will further evaluate teacher workload, student sentiment, and privacy clarity to refine pacing, prompts, and tracking controls. While challenges remain in usability and connectivity, MoodleQuest demonstrates a practical and privacy-aware approach to layering gamification onto Moodle while maintaining instructor oversight and institutional trust.

Keywords—Gamification, Asynchronous Learning, Learning Management Systems, Moodle, Student Engagement

I. INTRODUCTION

Asynchronous online learning has become a central mode of instruction in many schools and universities, yet students often struggle with motivation, sustained participation, and meaningful engagement when learning independently. While platforms such as Moodle provide structured content delivery, assignment workflows, and

basic activity logs, they offer limited mechanisms that support continuous motivation or provide clear indicators of learning effort. Without real-time feedback or interactive cues, students may treat coursework passively, leading to irregular study habits, reduced retention, and difficulty in measuring their own progress.

Gamification has emerged as a promising approach to address these issues. Research shows that visible rewards, timely engagement cues, and structured interactive elements can reinforce learning behaviors and improve completion rates in digital courses. However, implementing gamification directly within an LMS often requires heavy customization, which may create privacy concerns, administrative overhead, or disruptions to existing workflows.

MoodleQuest was developed to address these gaps by serving as a companion system that overlays gamified engagement and learning validation on top of Moodle through secure integration. Rather than replacing existing workflows, it adds quests, badges, experience points, leaderboards, virtual pets, and analytics that help students monitor their effort and progress. The system aims to provide instructors with clearer engagement insights while ensuring data transparency and user control. This paper presents the design, implementation, and evaluation plan for MoodleQuest as a practical and privacy-aware way to support motivation in asynchronous courses.

II. RELATED WORK

Prior research emphasizes the difficulty of sustaining engagement in asynchronous online courses. Wakjira and Bhattacharya [1] demonstrated that visual engagement indicators help instructors identify disengaged learners early. Kim et al. [2] showed that clustering techniques applied to activity data can more accurately characterize student behavior than basic participation counts. Ang et al.

[3], using the Community of Inquiry framework, highlighted the importance of structured interaction to support cognitive and social presence in asynchronous environments.

Other studies focus on feedback and analytics. Porter and Bozkaya [4] found that frequent or near-real-time feedback is associated with improved completion rates and learning outcomes. Doboli et al. [5] further demonstrated that multi-dimensional analytics incorporating cognitive, social, and emotional indicators can expose learning gaps that are not visible in standard LMS logs.

Gamification research aligns with these findings, showing that experience points, achievements, and challenge-based activities can enhance persistence and learner motivation. However, many existing systems rely on deep LMS modification or opaque AI-driven mechanisms with limited instructor oversight. MoodleQuest builds on prior work by emphasizing transparent, teacher-guided gamification layered beside Moodle rather than embedded within it.

Recent literature further supports the role of gamification and analytics in strengthening online learner engagement. Dichev and Dicheva [6] mapped current gamification practices in education and reported consistent motivational benefits when mechanics are aligned with the approach of teaching. Subhash and Cudney [7] found that gamified higher education environments frequently improve participation and persistence. Viberg et al. [8] described the expanding role of learning analytics in monitoring engagement and guiding instructional intervention. Martin et al. [9] showed that structured engagement strategies significantly influence learner satisfaction and participation in online courses. However, researchers also caution that poorly designed gamification can introduce negative behavioral effects, emphasizing the need for transparent and guided implementations [10].

III. PROBLEM STATEMENT

Although Moodle effectively manages content and assessments, it lacks mechanisms that sustain motivation and reveal real engagement in asynchronous settings. This creates several challenges:

A. *Passive participation*

Students may log in, open materials, or submit work without actively engaging or studying, making activity logs unreliable indicators of effort. Traditional metrics capture activity (clicks, views) but not engagement—sustained effort, reflection, or consistency—as shown by Wakjira and Bhattacharya who found visual logs alone insufficient for early detection. MoodleQuest counters this with behavioral proxies like streaks and quests that validate meaningful interaction.

B. *Limited feedback and motivation*

Asynchronous learning often lacks timely incentives or cues that help students maintain consistent study habits.

Without immediate rewards or progress signals, learners disengage, mirroring Porter and Bozkaya's findings on feedback delays, reducing completion rates. MoodleQuest introduces XP, badges, and daily quests for near-real-time reinforcement without altering core workflows.

C. *Insufficient visibility for teachers*

Educators have difficulty identifying disengaged students early because available analytics are minimal and do not capture behavioral patterns. Basic LMS reports overlook clustering patterns in activity data, as Kim et al. demonstrated with k-means analysis. MoodleQuest's heatmaps and early-warning flags provide multi-dimensional insights tied to teacher oversight.

D. *Lack of integrated gamification*

Moodle provides limited built-in gamification tools and customizing it may require technical expertise or compromise platform stability. Deep modifications risk data silos or security gaps, unlike lightweight companion approaches. MoodleQuest layers quests, leaderboards, and pets externally via secure APIs, preserving Moodle's integrity.

E. *Privacy and trust concerns*

Heavy tracking or automated scoring systems can create discomfort in students and increase administrative burdens. Opaque AI mechanisms erode trust, as noted in gamification critiques emphasizing transparency. MoodleQuest prioritizes teacher-guided validation, explicit consents, and granular controls to build institutional confidence.

MoodleQuest addresses these issues by providing a companion system that adds gamified engagement, clear progress indicators, and teacher-guided monitoring while maintaining privacy and seamless Moodle integration.

IV. SYSTEM DESIGN AND ARCHITECTURE

MoodleQuest follows a modular, two-tier architecture consisting of (1) a front-end engagement layer and (2) a middleware bridge that communicates with Moodle's web services, as shown in Fig. 1.

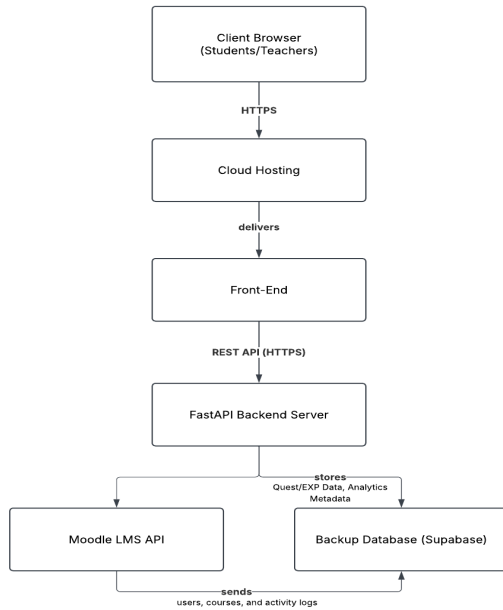


Figure 1. Deployment Architecture Diagram

Fig. 1 illustrates the deployment architecture and system data flow between the client browser, cloud hosting layer, middleware services, and Moodle integration endpoints.

A. Front-End Layer

The user interface is built with NextJS and TailwindCSS, providing responsive dashboards accessible on both mobile and desktop. As shown in Fig. 1, the front-end communicates securely with the backend through REST API calls over HTTPS. Features include:

- Gamification dashboard (EXP, badges, pet status)
- Quest and mission manager
- Leaderboards with filters and search
- Engagement heatmap and streak tracker
- Student and teacher analytics views

B. Middleware Layer

As illustrated in Fig. 1, a FastAPI service acts as a secure bridge between the front-end and Moodle:

- Handles authentication and session verification
- Uses role-based access control to manage student/teacher/admin permissions
- Fetches and updates Moodle data via REST web services
- Stores supplementary data (e.g., quest logs, streaks, badge events) in a backup database

C. Integration with Moodle

Based on the architecture shown in Fig. 1, MoodleQuest does not alter Moodle’s internal structure. Instead, it:

- Uses Moodle SSO for login
- Retrieves course data, activity completion, and grades through official APIs

- Syncs progress and engagement metrics without requiring duplicate accounts
- Respects Moodle’s permissions model to prevent unauthorized access

D. Non-Functional Considerations

Within the deployment model presented in Fig. 1, the system is designed with the following operational targets:

- Expected capacity: 100+ concurrent users
- Dashboard response target: under 5 seconds
- Encryption: TLS + Moodle’s built-in hashing mechanisms
- Privacy: granular tracking controls and explicit consent prompts

V. IMPLEMENTATION

Implementation followed an agile, sprint-based workflow with four main development phases:

A. Sprint 1 – Authentication and User Management

- Integration of Moodle SSO via token-based API
- Role-based access control for students, teachers, and administrators
- Profile synchronization and secure session handling

B. Sprint 2 – Core Gamification Features

- EXP engine with configurable reward rules
- Badge and achievement system
- Virtual pet evolution logic based on engagement
- Leaderboards with daily, weekly, and all-time filters

C. Sprint 3 – Quests and Progress Analytics

- Teacher tools for creating and assigning quests tied to learning objectives
- Daily quests and streak tracking
- Engagement heatmaps visualizing hourly and weekly activity
- Early-warning flags for low-activity students

D. Sprint 4 – Optimization and Testing

- Load testing for concurrency targets
- Refining chart rendering and caching for faster performance
- Usability walkthroughs with teachers for adjusting terminology, pacing, and privacy prompts
- Preparation for pilot deployment in a controlled classroom environment

VI. SURVEY METHODOLOGY AND DATA COLLECTION

A. Participants

A total of 32 students from Bankal National High School participated in the survey. All participants had prior experience using the MoodleQuest system during the pilot testing phase.

B. Survey Instrument

The evaluation employed a survey instrument consisting of 15 questions divided into two sections. The quantitative section included 12 Likert-scale items designed to measure three constructs: Perceived Engagement, Usability, and Learning Impact. The qualitative section consisted of three open-ended questions that gathered feedback on students' experiences, perceived difficulties, and suggested improvements to the system.

C. Measurement Scale

All quantitative items used a five-point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). Higher scores indicate a more positive perception of MoodleQuest. This scale was applied consistently across all quantitative items to allow comparison among the evaluated constructs.

D. Data Collection Procedure

The survey was conducted after the pilot deployment of MoodleQuest, allowing students to explore and interact with the system. Responses were collected electronically following system use. No personally identifiable information was collected, and responses were treated anonymously to encourage honest and unbiased feedback.

E. Data Analysis Approach

Quantitative survey data were analyzed using descriptive statistics, including mean scores, standard deviations, and the percentage of positive responses, defined as ratings of 4 (Agree) or 5 (Strongly Agree). Survey items were grouped according to the three predefined constructs. Qualitative responses were analyzed through thematic grouping, where recurring patterns and common themes were identified based on student feedback.

VII. RESULTS AND DISCUSSION

A. Overview of Survey Results

This section presents the results of the survey conducted to evaluate MoodleQuest as a gamified companion system for asynchronous learning. A total of 32 respondents completed 12 Likert-scale items, resulting in 384 quantitative responses. The overall mean score across all items was 4.16 with a standard deviation of 1.17, indicating a generally positive perception of the system. Approximately 64.8% of responses fell within the positive range (Agree to Strongly Agree), suggesting favorable acceptance among students.

Table I summarizes the consolidated quantitative results and dominant qualitative themes across the key evaluation dimensions of Perceived Engagement, Usability, Learning Impact, and Overall System Perception. As shown in Table I, Perceived Engagement and Learning Impact obtained the highest mean scores, while Usability exhibited greater variability, reflecting mixed user experiences with the system interface.

TABLE I. SUMMARY OF STUDENT PERCEPTIONS AND SYSTEM EVALUATION RESULTS

Evaluation	Quantitative Results (M, SD, % Positive)	Dominant Qualitative Themes	Interpretation
Perceived Engagement	4.38, 0.98, 71.1%	Motivation increase; progress visibility; reward satisfaction	Gamified elements supported sustained participation and task completion
Usability	3.53, 1.37, 44.5%	Navigation difficulty; feature overload; clarity issues	Interface complexity reduced ease of use for some learners
Learning Impact	4.58, 0.82, 78.9%	Habit formation; effort awareness; consistency	System supported self-regulated learning behaviors
Overall System Perception	4.16, 1.17, 64.8%	Positive acceptance; improvement requests	Motivational benefits outweighed usability limitations

B. Interpretation of Quantitative Findings

Analysis of individual survey items revealed consistently high ratings for learning-related outcomes. Items measuring improved study habits and overall learning impact received perfect mean scores, indicating unanimous agreement among respondents. Engagement-related items also scored highly, particularly those associated with motivation and enjoyment.

However, usability-related items showed greater variability in responses. While students reported ease of use and minimal need for assistance in some respects, lower scores related to navigation and feature clarity highlight inconsistencies in the user experience. These findings suggest that while MoodleQuest is effective in motivating learning behavior, its interface design and information architecture require refinement to improve accessibility and ease of use.

C. Qualitative Feedback Analysis

Qualitative feedback reinforced the quantitative results. Students highlighted gamification features such as badges, leaderboards, virtual pets, and progress tracking as the most engaging aspects of MoodleQuest. Reported challenges included navigation difficulties, occasional performance issues, unclear reward criteria, limited customization, and accessibility concerns. Suggested improvements focused on enhanced pet customization, integrated study tools, additional social features, and improved analytics, aligning with the identified usability issues.

D. Discussion

The findings of this study align with and extend prior research on engagement and motivation in asynchronous online learning environments. The high ratings for Perceived Engagement and Learning Impact support earlier work by Wakjira and Bhattacharya [1] and are further consistent with systematic reviews on educational gamification, which report motivational benefits when game mechanics are meaningfully aligned with instructional goals [6], [7]. In MoodleQuest, features such as quests, streaks, and progress visualization appear to function as engagement cues that encourage sustained participation, corroborating the conclusion that visible engagement indicators are critical for asynchronous settings.

The strong Learning Impact results further align with Porter and Bozkaya [4], who reported that timely feedback and reinforcement are associated with improved learning outcomes and course completion. These findings also reflect broader evidence that structured engagement strategies and feedback mechanisms significantly influence learner satisfaction and participation in online courses [9]. Although MoodleQuest does not deliver instructional feedback directly, its near-real-time rewards and progress indicators provide behavioral feedback that reinforces consistent study habits.

From an analytics perspective, the findings also resonate with Kim et al. [2] and are supported by research emphasizing the growing role of learning analytics in monitoring engagement and guiding instructional intervention [8]. MoodleQuest's use of streaks, quests, and aggregated engagement metrics supports the view that multi-dimensional indicators provide more meaningful representations of learner behavior than raw participation counts alone.

However, the lower Usability ratings highlight a limitation consistent with concerns raised in prior gamification and analytics research. Doboli et al. [5] noted that complex analytic systems risk increasing cognitive load when visualizations and features are not sufficiently intuitive. This concern is echoed in recent work cautioning that poorly designed gamification may introduce negative behavioral or motivational effects [10]. Similarly, qualitative feedback in this study indicates that navigation complexity and unclear feature affordances reduced ease of use for some students.

Importantly, MoodleQuest's companion-system approach distinguishes it from many gamified LMS implementations reported in the literature. Rather than deeply modifying the LMS core, the system reflects recommendations for transparent and guided gamification designs emphasized in recent gamification mappings and reviews [6], [7]. This design choice preserves Moodle's

native workflows and aligns with calls for greater transparency and instructor oversight in data-driven educational technologies.

Overall, the results suggest that MoodleQuest contributes to the existing body of research by demonstrating that analytics-informed and transparently implemented gamification can enhance engagement and perceived learning impact in asynchronous courses. At the same time, the findings reinforce prior evidence that usability remains a critical determinant of adoption and effectiveness in gamified and analytics-driven learning systems [8], [10].

VIII. CONCLUSION

MoodleQuest offers a practical and scalable approach to enhancing engagement in asynchronous learning by extending Moodle with transparent and teacher-guided gamification. By combining pets, quests, experience points, badges, leaderboards, and behavioral analytics, the system encourages consistent study habits and provides instructors with clearer insights into learner progress. Early evaluations indicate that streak of rewards and daily quests help students better understand their effort, while the integration approach maintains user trust by avoiding intrusive tracking and preserving Moodle's existing workflows.

Although challenges remain such as dependence on teacher validation and varying internet quality. The planned pilot studies aim to refine pacing, improve feedback loops, and assess the impact on student motivation. With further development in privacy-aware telemetry and objective-aligned recommendations, MoodleQuest has the potential to become a robust companion tool that strengthens learning outcomes in asynchronous environments.

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